

Лабораторная работа

Номер 15

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Получение навыков по работе с журналами системных событий

Создание файла конфигурации на сервере

```
[nsandryushin@server.nsandryushin.net ~]$ cd /etc/rsyslog.d  
[nsandryushin@server.nsandryushin.net rsyslog.d]$ sudo touch netlog-server.conf  
[sudo] password for nsandryushin:  
[nsandryushin@server.nsandryushin.net rsyslog.d]$ sudo nano netlog-server.conf
```

Рис. 1: Создание файла конфигурации на сервере

Редактирование конфигурации rsyslog на сервере

A screenshot of the GNU nano 8.1 text editor. The title bar at the top shows 'GNU nano 8.1' on the left and 'netlog-server.conf' on the right. The editor area contains three lines of text: '\$ModLoad imtcp', '\$InputTCPServerRun 514', and a blank line. The cursor is positioned at the end of the third line.

```
GNU nano 8.1 netlog-server.conf
$ModLoad imtcp
$InputTCPServerRun 514
```

Рис. 2: Редактирование конфигурации rsyslog на сервере

Перезапуск службы и начало проверки портов

```
[nsandryushin@server.nsandryushin.net rsyslog.d]$ sudo systemctl restart rsyslog
[nsandryushin@server.nsandryushin.net rsyslog.d]$ sudo lsof | grep TCP
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1001/gvfs
Output information may be incomplete.
lsof: WARNING: can't stat() fuse.portal file system /run/user/1001/doc
Output information may be incomplete.
systemd      1                root    42u     IPv4            6235      0t0      TCP *:sun
rpc (LISTEN)
systemd      1                root    44u     IPv6            6247      0t0      TCP *:sun
rpc (LISTEN)
systemd      1                root    91u     IPv6            9058      0t0      TCP *:web
sm (LISTEN)
rpcbind      861              rpc      5u     IPv4            6235      0t0      TCP *:sun
rpc (LISTEN)
rpcbind      861              rpc      7u     IPv6            6247      0t0      TCP *:sun
rpc (LISTEN)
cupsd        1264             root     7u     IPv6            11459     0t0      TCP local
host:ipp (LISTEN)
cupsd        1264             root     8u     IPv4            11460     0t0      TCP local
host:ipp (LISTEN)
sshd         1274             root     7u     IPv4            9903      0t0      TCP *:dow
n (LISTEN)
sshd         1274             root     8u     IPv6            9905      0t0      TCP *:dow
n (LISTEN)
sshd         1274             root     9u     IPv4            9907      0t0      TCP *:ssh
(LISTEN)
sshd         1274             root    10u     IPv6            9909      0t0      TCP *:ssh
(LISTEN)
named        1377             named    27u     IPv4            11674     0t0      TCP local
host:domain (LISTEN)
named        1377             named    28u     IPv4            11675     0t0      TCP local
```

Рис. 3: Перезапуск службы и начало проверки портов

Подтверждение прослушивания порта 514

```
rsyslogd 8324 root 4u IPv4 40059 0t0 TCP *:she
ll (LISTEN) root 5u IPv6 40060 0t0 TCP *:she
rsyslogd 8324 8327 in:imjour root 4u IPv4 40059 0t0 TCP *:she
ll (LISTEN) rsyslogd 8324 8327 in:imjour root 5u IPv6 40060 0t0 TCP *:she
ll (LISTEN) rsyslogd 8324 8328 in:imtcp root 4u IPv4 40059 0t0 TCP *:she
ll (LISTEN) rsyslogd 8324 8328 in:imtcp root 5u IPv6 40060 0t0 TCP *:she
ll (LISTEN) rsyslogd 8324 8329 in:imtcp root 4u IPv4 40059 0t0 TCP *:she
ll (LISTEN) rsyslogd 8324 8329 in:imtcp root 5u IPv6 40060 0t0 TCP *:she
ll (LISTEN) rsyslogd 8324 8330 in:imtcp root 4u IPv4 40059 0t0 TCP *:she
ll (LISTEN) rsyslogd 8324 8330 in:imtcp root 5u IPv6 40060 0t0 TCP *:she
ll (LISTEN) rsyslogd 8324 8331 in:imtcp root 4u IPv4 40059 0t0 TCP *:she
ll (LISTEN) rsyslogd 8324 8331 in:imtcp root 5u IPv6 40060 0t0 TCP *:she
ll (LISTEN) rsyslogd 8324 8332 in:imtcp root 4u IPv4 40059 0t0 TCP *:she
ll (LISTEN) rsyslogd 8324 8332 in:imtcp root 5u IPv6 40060 0t0 TCP *:she
ll (LISTEN) rsyslogd 8324 8335 rs:main root 4u IPv4 40059 0t0 TCP *:she
ll (LISTEN) rsyslogd 8324 8335 rs:main root 5u IPv6 40060 0t0 TCP *:she
ll (LISTEN)
[nsandryushin@server.nsandryushin.net rsyslog.d]$ █
```

Рис. 4: Подтверждение прослушивания порта 514

Настройка firewall на сервере

```
[nsandryushin@server.nsandryushin.net rsyslog.d]$ sudo firewall-cmd --add-port=514/tcp  
success  
[nsandryushin@server.nsandryushin.net rsyslog.d]$ sudo firewall-cmd --add-port=514/tcp --permanent  
success  
[nsandryushin@server.nsandryushin.net rsyslog.d]$ █
```

Рис. 5: Настройка firewall на сервере

Создание файла конфигурации на клиенте

```
[nsandryushin@client.nsandryushin.net ~]$ cd /etc/rsyslog.d
[nsandryushin@client.nsandryushin.net rsyslog.d]$ sudo netlog-client.conf
[sudo] password for nsandryushin:
sudo: netlog-client.conf: command not found
[nsandryushin@client.nsandryushin.net rsyslog.d]$ sudo touch netlog-client.conf
[nsandryushin@client.nsandryushin.net rsyslog.d]$ nano netlog-client.conf █
```

Рис. 6: Создание файла конфигурации на клиенте

Редактирование конфигурации rsyslog на клиенте



```
GNU nano 8.1 netlog-client.conf
*.* @@server.nsandryushin.net:514
```

Рис. 7: Редактирование конфигурации rsyslog на клиенте

Просмотр логов на сервере через tail

```
[nsandryushin@server nsandryushin.net rsyslog.d]$ sudo tail -f /var/log/messages
Dec 13 18:31:39 server systemd[1]: Stopped user-runtime-dir@0.service - User Runtime Directory /run/user/0.
Dec 13 18:31:39 server systemd[1]: Removed slice user-0.slice - User Slice of UID 0.
Dec 13 18:33:20 server systemd-logind[924]: Existing logind session ID 3 used by new audit session, ignoring.
Dec 13 18:33:20 server systemd[1]: Created slice user-0.slice - User Slice of UID 0.
Dec 13 18:33:20 server systemd[1]: Starting user-runtime-dir@0.service - User Runtime Directory /run/user/0.
Dec 13 18:33:20 server systemd-logind[924]: New session c8 of user root.
Dec 13 18:33:20 server systemd[1]: Finished user-runtime-dir@0.service - User Runtime Directory /run/user/0.
Dec 13 18:33:20 server systemd[1]: Starting user@0.service - User Manager for UID 0...
Dec 13 18:33:20 server systemd-logind[924]: New session 7 of user root.
Dec 13 18:33:20 server systemd[8447]: Queued start job for default target default.target.
Dec 13 18:33:20 server systemd[8447]: Created slice app.slice - User Application Slice.
Dec 13 18:33:20 server systemd[8447]: grub-boot-success.timer - Mark boot as successful after the user session has run 2 minutes was skipped because of an unset condition check (ConditionUser=!@system).
Dec 13 18:33:20 server systemd[8447]: Started systemd-tmpfiles-clean.timer - Daily Cleanup of User's Temporary Directories.
Dec 13 18:33:20 server systemd[8447]: Reached target paths.target - Paths.
Dec 13 18:33:20 server systemd[8447]: Reached target timers.target - Timers.
Dec 13 18:33:20 server systemd[8447]: Starting dbus.socket - D-Bus User Message Bus Socket...
Dec 13 18:33:20 server systemd[8447]: pipewire-pulse.socket - PipeWire PulseAudio was skipped because of an unset condition check (ConditionUser!=root).
Dec 13 18:33:20 server systemd[8447]: Listening on pipewire.socket - PipeWire Multimedia System Sockets.
Dec 13 18:33:20 server systemd[8447]: Starting systemd-tmpfiles-setup.service - Create User Files and Directories...
Dec 13 18:33:20 server systemd[8447]: Finished systemd-tmpfiles-setup.service - Create User Files and Directories.
Dec 13 18:33:20 server systemd[8447]: Listening on dbus.socket - D-Bus User Message Bus Socket.
Dec 13 18:33:20 server systemd[8447]: Reached target sockets.target - Sockets.
Dec 13 18:33:20 server systemd[8447]: Reached target basic.target - Basic System.
Dec 13 18:33:20 server systemd[8447]: Reached target default.target - Main User Target.
Dec 13 18:33:20 server systemd[8447]: Startup finished in 259ms.
Dec 13 18:33:20 server systemd[1]: Started user@0.service - User Manager for UID 0.
Dec 13 18:33:20 server systemd[1]: Started session-c8.scope - Session c8 of User root.
```

Рис. 8: Просмотр логов на сервере через tail

Интерфейс Gnome System Monitor

nsandryushinserver.nsandryushin.net syslog.d[5] gnome-system-monitor									
Processes Resources File Systems									
Process Name	User	% CPU	IO	Memory	Disk read rate	Disk write rate	Disk read	Disk write	Priority
at-spi2-registr	nsandryushin	0.00	5416	321.1 kB	705.6 kB/s	N/A	N/A	N/A	N/A
at-spi2-bus-launcher	nsandryushin	0.00	6486	262.1 kB	667.6 kB/s	N/A	N/A	N/A	N/A
bash	nsandryushin	0.00	7887	3.5 kB	9.8 kB/s	N/A	N/A	N/A	N/A
bash	nsandryushin	0.00	8477	2.0 MB	843.8 kB/s	N/A	N/A	N/A	N/A
catatone	nsandryushin	0.00	7843	N/A	653.6 kB/s	N/A	N/A	N/A	N/A
dbus-broker	nsandryushin	0.00	5287	1.3 MB	2.1 MB/s	N/A	N/A	N/A	N/A
dbus-broker	nsandryushin	0.00	6415	131.1 kB	184.3 kB/s	N/A	N/A	N/A	N/A
dbus-broker-launch	nsandryushin	0.00	5296	N/A	254.8 kB/s	N/A	N/A	N/A	N/A
dbus-launch	nsandryushin	0.00	6411	N/A	N/A	N/A	N/A	N/A	N/A
dnscd-service	nsandryushin	0.00	5448	392.0 kB	479.2 kB/s	634.4 kB/s	N/A	N/A	N/A
evolution-addressbook-factory	nsandryushin	0.00	6782	131.1 kB	3.8 MB/s	53.2 kB/s	N/A	N/A	N/A
evolution-alarm-notify	nsandryushin	0.00	5584	262.1 kB	4.5 MB/s	N/A	N/A	N/A	N/A
evolution-calendar-factory	nsandryushin	0.00	6522	93.7 kB	4.3 MB/s	N/A	N/A	N/A	N/A
evolution-source-registry	nsandryushin	0.00	6454	131.1 kB	3.8 MB/s	N/A	N/A	N/A	N/A
firefox	nsandryushin	4.54	7118	126.3 MB	405.8 kB/s	49.6 MB/s	27.8 kB/s	N/A	N/A
gnome-keyring-session	nsandryushin	0.00	6795	N/A	20.5 MB/s	N/A	N/A	N/A	N/A
gnome	nsandryushin	0.00	6468	32.8 kB	2.3 MB/s	N/A	N/A	N/A	N/A
gnome	nsandryushin	0.00	6624	3.3 MB	5.2 MB/s	N/A	N/A	N/A	N/A
gnome-keyring-daemon	nsandryushin	0.00	6269	322.6 kB	2.0 MB/s	4.1 kB/s	N/A	N/A	N/A
gnome-session-binary	nsandryushin	0.00	5289	131.1 kB	526.2 kB/s	N/A	N/A	N/A	N/A
gnome-session-binary	nsandryushin	0.00	6342	655.4 kB	3.2 MB/s	4.1 kB/s	N/A	N/A	N/A
gnome-session-clib	nsandryushin	0.00	6349	N/A	24.6 kB/s	N/A	N/A	N/A	N/A
gnome-shell	nsandryushin	21.22	6309	280.1 kB	322.3 kB/s	139.3 kB/s	1.3 kB/s	N/A	N/A
gnome-shell	nsandryushin	0.00	6432	786.2 kB	5.4 MB/s	N/A	N/A	N/A	N/A
gnome-software	nsandryushin	0.00	6528	6.8 MB	68.6 MB/s	N/A	N/A	N/A	N/A
gnome-system-monitor	nsandryushin	5.05	5586	144.9 MB	48.6 MB/s	53.2 kB/s	N/A	N/A	N/A
gnome-terminal	nsandryushin	0.00	5587	131.1 kB	639.5 kB/s	N/A	N/A	N/A	N/A
gnome-identity-service	nsandryushin	0.00	6691	131.1 kB	397.3 kB/s	N/A	N/A	N/A	N/A
gnome-id3-settings	nsandryushin	0.00	6465	131.1 kB	323.5 kB/s	N/A	N/A	N/A	N/A
gnome-cd	nsandryushin	0.00	6466	262.1 kB	978.0 kB/s	N/A	N/A	N/A	N/A
gnome-datetime	nsandryushin	0.00	6475	131.1 kB	387.2 kB/s	N/A	N/A	N/A	N/A
gnome-disk-util-notify	nsandryushin	0.00	5586	131.1 kB	159.7 kB/s	N/A	N/A	N/A	N/A
gnome-housekeeping	nsandryushin	0.00	6483	392.2 kB	647.2 kB/s	N/A	N/A	N/A	N/A
gnome-keyboard	nsandryushin	0.00	6484	262.1 kB	1.8 MB/s	N/A	N/A	N/A	N/A
gnome-media-libs	nsandryushin	0.00	6485	93.7 kB	1.8 MB/s	N/A	N/A	N/A	N/A

Рис. 9: Интерфейс Gnome System Monitor

Ручная установка lnav на сервере

```
[nsandryushin@server.nsandryushin.net rsyslog.d]$ cd ~/Downloads/  
[nsandryushin@server.nsandryushin.net Downloads]$ sudo cp lnav /usr/bin  
[nsandryushin@server.nsandryushin.net Downloads]$
```

Рис. 10: Ручная установка lnav на сервере

Просмотр логов через lnav на сервере

```
2015-12-17T18:38:21.000  syslog.log  messages[6050]  system[1] 1
Dec 13 18:38:18 server systemd[6447]: Stopped system-tmpfiles-clean.timer - Daily Cleanup of User's T
Dec 13 18:38:18 server systemd[6447]: Closed dbus.socket - D-Bus User Message Bus Socket.
Dec 13 18:38:18 server systemd[6447]: Closed pipewire.socket - PipeWire Multimedia System Sockets.
Dec 13 18:38:18 server systemd[6447]: Stopped system-tmpfiles-setup.service - Create User Files and Di
Dec 13 18:38:18 server systemd[6447]: Removed elice.app.slice - User Application Slice.
Dec 13 18:38:18 server systemd[6447]: Reached target shutdown.target - Shutdown.
Dec 13 18:38:18 server systemd[6447]: Finished system-exit.service - Exit the Session.
Dec 13 18:38:18 server systemd[6447]: Reached target exit.target - Exit the Session.
Dec 13 18:38:18 server systemd-logind[524]: Removed session 7.
Dec 13 18:38:18 server systemd[1]: user@service: Deactivated successfully.
Dec 13 18:38:18 server systemd[1]: Stopped user@service - User Manager for UID 0.
Dec 13 18:38:18 server systemd[1]: Stopping user-runtime-dirs@service - User Runtime Directory /run/us
Dec 13 18:38:18 server systemd[1]: run-user-0.mount: Deactivated successfully.
Dec 13 18:38:18 server systemd[1]: user-runtime-dirs@service: Deactivated successfully.
Dec 13 18:38:18 server systemd[1]: Stopped user-runtime-dirs@service - User Runtime Directory /run/us
Dec 13 18:38:18 server systemd[1]: Removed slice user-0.slice - User Slice of UID 0.
Dec 13 18:38:18 server systemd[1]: user-0.slice: Consumed 1.465s CPU time, 204.3M memory peak, 3.8M mem
Dec 13 18:38:20 server systemd-logind[524]: Existing login session ID 3 used by new audit session, ign
Dec 13 18:38:20 server systemd[1]: Created slice user-0.slice - User Slice of UID 0.
Dec 13 18:38:20 server systemd[1]: Starting user-runtime-dirs@service - User Runtime Directory /run/us
Dec 13 18:38:20 server systemd-logind[524]: New session cll of user root.
Dec 13 18:38:20 server systemd[1]: Finished user-runtime-dirs@service - User Runtime Directory /run/us
Dec 13 18:38:20 server systemd[1]: Starting user@service - User Manager for UID 0...
Dec 13 18:38:20 server systemd-logind[524]: New session 0 of user root.
Dec 13 18:38:21 server systemd[10496]: Owned start job for default target default.target.
Dec 13 18:38:21 server systemd[10496]: Created slice app.slice - User Application Slice.
Dec 13 18:38:21 server systemd[10496]: gnu-boot-success.timer - Mark boot as successful after the use
Dec 13 18:38:21 server systemd[10496]: Started system-tmpfiles-clean.timer - Daily Cleanup of User's T
Dec 13 18:38:21 server systemd[10496]: Reached target paths.target - Paths.
Dec 13 18:38:21 server systemd[10496]: Reached target clocks.target - Times.
Dec 13 18:38:21 server systemd[10496]: Starting dbus.socket - D-Bus User Message Bus Socket...
Dec 13 18:38:21 server systemd[10496]: pipewire-pulse.socket - PipeWire PulseAudio was skipped because
Dec 13 18:38:21 server systemd[10496]: Listening on pipewire.socket - PipeWire Multimedia System Socket
Dec 13 18:38:21 server systemd[10496]: Starting system-tmpfiles-setup.service - Create User Files and
Dec 13 18:38:21 server systemd[10496]: Listening on dbus.socket - D-Bus User Message Bus Socket.
Dec 13 18:38:21 server systemd[10496]: Finished system-tmpfiles-setup.service - Create User Files and
Dec 13 18:38:21 server systemd[10496]: Reached target sockets.target - Sockets.
Dec 13 18:38:21 server systemd[10496]: Reached target basic.target - Basic System.
Dec 13 18:38:21 server systemd[10496]: Reached target default.target - Main User Target.
Dec 13 18:38:21 server systemd[10496]: Startup finished in 106ms.
Dec 13 18:38:21 server systemd[1]: Started user@service - User Manager for UID 0.
Dec 13 18:38:21 server systemd[1]: Started session-cll.scope - Session cll of User Root.
```

Рис. 11: Просмотр логов через lnav на сервере

Установка и запуск lnav на клиенте

```
[nsandryushin@client.nsandryushin.net client]$ cd ~/Downloads/  
[nsandryushin@client.nsandryushin.net Downloads]$ ls  
015-netlog.pdf  lnav  lnav-0.13.2-linux-musl-x86_64  lnav-0.13.2-linux-musl-x86_64.zip  
[nsandryushin@client.nsandryushin.net Downloads]$ sudo cp lnav /usr/bin  
[sudo] password for nsandryushin:  
[nsandryushin@client.nsandryushin.net Downloads]$ sudo lnav
```

Рис. 12: Установка и запуск lnav на клиенте

Просмотр логов через Inav на клиенте

```
2025-12-13T18:49:09 UTC                                     Press F2 to enable mouse support
10:00 2025-12-13T18:49:07:000 : syslog_log : messages[9647] : systemd[1] :
Dec 13 18:48:49 client systemd[5028]: Stopped systemd-tmpfiles-clean.timer - Daily Cleanup of User's Te
Dec 13 18:48:49 client systemd[5028]: Closed dbus.socket - D-Bus User Message Bus Socket.
Dec 13 18:48:49 client systemd[5028]: Closed pipewire.socket - PipeWire Multimedia System Sockets.
Dec 13 18:48:49 client systemd[5028]: Stopped systemd-tmpfiles-setup.service - Create User Files and Di
Dec 13 18:48:49 client systemd[5028]: Removed alice app.slice - User Application Slice.
Dec 13 18:48:49 client systemd[5028]: Reached target shutdown.target - Shutdown.
Dec 13 18:48:49 client systemd[5028]: Finished systemd-exit.service - Exit the Session.
Dec 13 18:48:49 client systemd[5028]: Reached target exit.target - Exit the Session.
Dec 13 18:48:50 client systemd-logind[902]: Removed session 11.
Dec 13 18:48:50 client systemd[1]: user@.service: Deactivated successfully.
Dec 13 18:48:50 client systemd[1]: Stopped user@.service - User Manager for UID 0.
Dec 13 18:48:50 client systemd[1]: Stopping user-runtime-dir@.service - User Runtime Directory /run/us
Dec 13 18:48:50 client systemd[1]: run-user@.mount: Deactivated successfully.
Dec 13 18:48:50 client systemd[1]: user-runtime-dir@.service: Deactivated successfully.
Dec 13 18:48:50 client systemd[1]: Stopped user-runtime-dir@.service - User Runtime Directory /run/us
Dec 13 18:48:50 client systemd[1]: Removed alice user@.alice - User Slice of UID 0.
Dec 13 18:48:50 client systemd[1]: user@.alice: Consumed 214ms CPU time, 104.6M memory peak.
Dec 13 18:49:07 client systemd-logind[902]: Existing logind session ID 3 used by new audit session, ign
Dec 13 18:49:07 client systemd-logind[902]: New session c11 of user root.
Dec 13 18:49:07 client systemd[1]: Created alice user@.alice - User Slice of UID 0.
Dec 13 18:49:07 client systemd[1]: Starting user-runtime-dir@.service - User Runtime Directory /run/us
Dec 13 18:49:07 client systemd[1]: Finished user-runtime-dir@.service - User Runtime Directory /run/us
Dec 13 18:49:07 client systemd[1]: Starting user@.service - User Manager for UID 0...
Dec 13 18:49:07 client systemd-logind[902]: New session 12 of user root.
Dec 13 18:49:07 client systemd[5009]: Queued start job for default target default.target.
Dec 13 18:49:07 client systemd[5009]: Created alice app.slice - User Application Slice.
Dec 13 18:49:07 client systemd[5009]: grub-boot-success.timer - Mark boot as successful after the user
Dec 13 18:49:07 client systemd[5009]: Started systemd-tmpfiles-clean.timer - Daily Cleanup of User's Te
Dec 13 18:49:07 client systemd[5009]: Reached target paths.target - Paths.
Dec 13 18:49:07 client systemd[5009]: Reached target timers.target - Timers.
Dec 13 18:49:07 client systemd[5009]: Starting dbus.socket - D-Bus User Message Bus Socket...
Dec 13 18:49:07 client systemd[5009]: pipewire-pulse.socket - PipeWire PulseAudio was skipped because o
Dec 13 18:49:07 client systemd[5009]: Listening on pipewire.socket - PipeWire Multimedia System Sockets
Dec 13 18:49:07 client systemd[5009]: Starting systemd-tmpfiles-setup.service - Create User Files and D
Dec 13 18:49:07 client systemd[5009]: Finished systemd-tmpfiles-setup.service - Create User Files and D
Dec 13 18:49:07 client systemd[5009]: Listening on dbus.socket - D-Bus User Message Bus Socket.
Dec 13 18:49:07 client systemd[5009]: Reached target sockets.target - Sockets.
Dec 13 18:49:07 client systemd[5009]: Reached target basic.target - Basic System.
Dec 13 18:49:07 client systemd[5009]: Reached target default.target - Main User Target.
Dec 13 18:49:07 client systemd[5009]: Startup finished in 164ms.
Dec 13 18:49:07 client systemd[1]: Started user@.service - User Manager for UID 0.
Dec 13 18:49:07 client systemd[1]: Started session-c11.scope - Session c11 of User root.
```

Рис. 13: Просмотр логов через Inav на клиенте

Подготовка файлов для автоматизации сервера

```
[nsandryushin@server.nsandryushin.net rsyslog.d]$ cd /vagrant/provision/server
[nsandryushin@server.nsandryushin.net server]$ sudo mkdir -p /vagrant/provision/server/netlog/etc/rsyslog
.d
[sudo] password for nsandryushin:
[nsandryushin@server.nsandryushin.net server]$ sudo cp -R /etc/rsyslog.d/netlog-server.conf /vagrant/prov
ision/server/netlog/etc/rsyslog.d
[nsandryushin@server.nsandryushin.net server]$ cd /vagrant/provision/server
[nsandryushin@server.nsandryushin.net server]$ sudo touch netlog.sh
[nsandryushin@server.nsandryushin.net server]$ sudo nano netlog.sh
[nsandryushin@server.nsandryushin.net server]$ sudo chmod +x netlog.sh
[nsandryushin@server.nsandryushin.net server]$ █
```

Рис. 14: Подготовка файлов для автоматизации сервера

Скрипт provisioning для сервера

```
GNU nano 8.1                                     netlog.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Copy configuration files"
cp -R /vagrant/provision/server/netlog/etc/* /etc
restorecon -vR /etc
echo "Configure firewall"
firewall-cmd --add-port=514/tcp
firewall-cmd --add-port=514/tcp --permanent
echo "Start rsyslog service"
systemctl restart rsyslog
```

Рис. 15: Скрипт provisioning для сервера

Подготовка файлов для автоматизации клиента

```
[nsandryushin@client.nsandryushin.net rsyslog.d]$ cd /vagrant/provision/client
[nsandryushin@client.nsandryushin.net client]$ sudo mkdir -p /vagrant/provision/client/netlog/etc/rsyslog
.d
[sudo] password for nsandryushin:
[nsandryushin@client.nsandryushin.net client]$ sudo cp -R /etc/rsyslog.d/netlog-client.conf
cp: missing destination file operand after '/etc/rsyslog.d/netlog-client.conf'
Try 'cp --help' for more information.
[nsandryushin@client.nsandryushin.net client]$ sudo cp -R /etc/rsyslog.d/netlog-client.conf /vagrant/prov
ision/client/netlog/etc/rsyslog.d/
[nsandryushin@client.nsandryushin.net client]$ cd /vagrant/provision/client
[nsandryushin@client.nsandryushin.net client]$ sudo touch netlog.sh
[nsandryushin@client.nsandryushin.net client]$ sudo chmod +x netlog.sh
[nsandryushin@client.nsandryushin.net client]$ sudo nano netlog.sh
[nsandryushin@client.nsandryushin.net client]$ █
```

Рис. 16: Подготовка файлов для автоматизации клиента

Скрипт provisioning для клиента

```
GNU nano 8.1 netlog.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install lnav
echo "Copy configuration files"
cp -R /vagrant/provision/client/netlog/etc/* /etc
restorecon -vR /etc
echo "Start rsyslog service"
systemctl restart rsyslog
```

Рис. 17: Скрипт provisioning для клиента

Настройка Vagrantfile

```

120         username="${user}"
121         password="${password}/ssh.sh"
122
123     server.vw.provision
124         "server/setting",
125         type="shell",
126         provision_order: true,
127         path: "${provision/client/setting.sh}"
128
129
130     and
131
132     # client configuration
133     client.vw.define "client", antipostart: false do |client|
134         client.vw.box = "ubuntu1204"
135         client.vw.hostname = "client"
136
137         client.vw.boot_timeout = 1440
138
139         client.vw.insert_key = false
140         client.vw.username = "vagrant"
141         client.vw.password = "vagrant"
142
143         client.vw.network | private_network,
144             type: "dhcp",
145             virtualbox____interface: "vnet1"
146
147         client.vw.provider__virtualbox__virtualbox__
148             virtualbox______name__ = "Vagrant__", ____order__, ____m__
149             virtualbox______customize__ "modify", __id__, ____redpack__, ____box__
150
151     and
152
153     client.vw.provision "client_dummy",
154         type: "shell",
155         provision_order: true,
156         path: "${provision/client/01-dummy.sh}"
157
158     client.vw.provision "client_setting",
159         type: "shell",
160         provision_order: true,
161         path: "${provision/client/01-setting.sh}"
162
163     client.vw.provision "client_mail",
164         type: "shell",
165         provision_order: true,
166         path: "${provision/client/mail.sh}"
167
168     client.vw.provision "client_ntp",
169         type: "shell",
170         provision_order: true,
171         path: "${provision/client/ntp.sh}"
172
173     client.vw.provision "client_ntp",
174         type: "shell",
175         provision_order: true,
176         path: "${provision/client/ntp.sh}"
177
178     client.vw.provision "client_ntp",
179         type: "shell",
180         provision_order: true,
181         path: "${provision/client/ntp.sh}"
182
183     client.vw.provision "client_ntp",
184         type: "shell",
185         provision_order: true,
186         path: "${provision/client/ntp.sh}"
187
188     client.vw.provision "client_ntp",
189         type: "shell",
190         provision_order: true,
191         path: "${provision/client/ntp.sh}"
192
193     client.vw.provision "client_ntp",
194         type: "shell",
195         provision_order: true,
196         path: "${provision/client/ntp.sh}"
197
198     client.vw.provision "client_ntp",
199         type: "shell",
200         provision_order: true,
201         path: "${provision/client/ntp.sh}"
202
203     client.vw.provision "client_ntp",
204         type: "shell",
205         provision_order: true,
206         path: "${provision/client/ntp.sh}"
207
208     client.vw.provision "client_ntp",
209         type: "shell",
210         provision_order: true,
211         path: "${provision/client/ntp.sh}"
212
213     client.vw.provision "client_ntp",
214         type: "shell",
215         provision_order: true,
216         path: "${provision/client/ntp.sh}"
217
218     client.vw.provision "client_ntp",
219         type: "shell",
220         provision_order: true,
221         path: "${provision/client/ntp.sh}"
222
223     client.vw.provision "client_ntp",
224         type: "shell",
225         provision_order: true,
226         path: "${provision/client/ntp.sh}"
227
228     client.vw.provision "client_ntp",
229         type: "shell",
230         provision_order: true,
231         path: "${provision/client/ntp.sh}"
232
233     client.vw.provision "client_ntp",
234         type: "shell",
235         provision_order: true,
236         path: "${provision/client/ntp.sh}"
237
238     client.vw.provision "client_ntp",
239         type: "shell",
240         provision_order: true,
241         path: "${provision/client/ntp.sh}"
242
243     client.vw.provision "client_ntp",
244         type: "shell",
245         provision_order: true,
246         path: "${provision/client/ntp.sh}"
247
248     client.vw.provision "client_ntp",
249         type: "shell",
250         provision_order: true,
251         path: "${provision/client/ntp.sh}"
252
253     client.vw.provision "client_ntp",
254         type: "shell",
255         provision_order: true,
256         path: "${provision/client/ntp.sh}"
257
258     client.vw.provision "client_ntp",
259         type: "shell",
260         provision_order: true,
261         path: "${provision/client/ntp.sh}"
262
263     client.vw.provision "client_ntp",
264         type: "shell",
265         provision_order: true,
266         path: "${provision/client/ntp.sh}"
267
268     client.vw.provision "client_ntp",
269         type: "shell",
270         provision_order: true,
271         path: "${provision/client/ntp.sh}"
272
273     client.vw.provision "client_ntp",
274         type: "shell",
275         provision_order: true,
276         path: "${provision/client/ntp.sh}"
277
278     client.vw.provision "client_ntp",
279         type: "shell",
280         provision_order: true,
281         path: "${provision/client/ntp.sh}"
282
283     client.vw.provision "client_ntp",
284         type: "shell",
285         provision_order: true,
286         path: "${provision/client/ntp.sh}"
287
288     client.vw.provision "client_ntp",
289         type: "shell",
290         provision_order: true,
291         path: "${provision/client/ntp.sh}"
292
293     client.vw.provision "client_ntp",
294         type: "shell",
295         provision_order: true,
296         path: "${provision/client/ntp.sh}"
297
298     client.vw.provision "client_ntp",
299         type: "shell",
300         provision_order: true,
301         path: "${provision/client/ntp.sh}"
302
303     client.vw.provision "client_ntp",
304         type: "shell",
305         provision_order: true,
306         path: "${provision/client/ntp.sh}"
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308     client.vw.provision "client_ntp",
309         type: "shell",
310         provision_order: true,
311         path: "${provision/client/ntp.sh}"
312
313     client.vw.provision "client_ntp",
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315         provision_order: true,
316         path: "${provision/client/ntp.sh}"
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318     client.vw.provision "client_ntp",
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321         path: "${provision/client/ntp.sh}"
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323     client.vw.provision "client_ntp",
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326         path: "${provision/client/ntp.sh}"
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336         path: "${provision/client/ntp.sh}"
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346         path: "${provision/client/ntp.sh}"
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348     client.vw.provision "client_ntp",
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350         provision_order: true,
351         path: "${provision/client/ntp.sh}"
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355         provision_order: true,
356         path: "${provision/client/ntp.sh}"
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366         path: "${provision/client/ntp.sh}"
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371         path: "${provision/client/ntp.sh}"
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376         path: "${provision/client/ntp.sh}"
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378     client.vw.provision "client_ntp",
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380         provision_order: true,
381         path: "${provision/client/ntp.sh}"
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383     client.vw.provision "client_ntp",
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385         provision_order: true,
386         path: "${provision/client/ntp.sh}"
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388     client.vw.provision "client_ntp",
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391         path: "${provision/client/ntp.sh}"
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394         type: "shell",
395         provision_order: true,
396         path: "${provision/client/ntp.sh}"
397
398     client.vw.provision "client_ntp",
399         type: "shell",
400         provision_order: true,
401         path: "${provision/client/ntp.sh}"
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403     client.vw.provision "client_ntp",
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405         provision_order: true,
406         path: "${provision/client/ntp.sh}"
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408     client.vw.provision "client_ntp",
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411         path: "${provision/client/ntp.sh}"
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413     client.vw.provision "client_ntp",
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415         provision_order: true,
416         path: "${provision/client/ntp.sh}"
417
418     client.vw.provision "client_ntp",
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420         provision_order: true,
421         path: "${provision/client/ntp.sh}"
422
423     client.vw.provision "client_ntp",
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425         provision_order: true,
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427
428     client.vw.provision "client_ntp",
429         type: "shell",
430         provision_order: true,
431         path: "${provision/client/ntp.sh}"
432
433     client.vw.provision "client_ntp",
434         type: "shell",
435         provision_order: true,
436         path: "${provision/client/ntp.sh}"
437
438     client.vw.provision "client_ntp",
439         type: "shell",
440         provision_order: true,
441         path: "${provision/client/ntp.sh}"
442
443     client.vw.provision "client_ntp",
444         type: "shell",
445         provision_order: true,
446         path: "${provision/client/ntp.sh}"
447
448     client.vw.provision "client_ntp",
449         type: "shell",
450         provision_order: true,
451         path: "${provision/client/ntp.sh}"
452
453     client.vw.provision "client_ntp",
454         type: "shell",
455         provision_order: true,
456         path: "${provision/client/ntp.sh}"
457
458     client.vw.provision "client_ntp",
459         type: "shell",
460         provision_order: true,
461         path: "${provision/client/ntp.sh}"
462
463     client.vw.provision "client_ntp",
464         type: "shell",
465         provision_order: true,
466         path: "${provision/client/ntp.sh}"
467
468     client.vw.provision "client_ntp",
469         type: "shell",
470         provision_order: true,
471         path: "${provision/client/ntp.sh}"
472
473     client.vw.provision "client_ntp",
474         type: "shell",
475         provision_order: true,
476         path: "${provision/client/ntp.sh}"
477
478     client.vw.provision "client_ntp",
479         type: "shell",
480         provision_order: true,
481         path: "${provision/client/ntp.sh}"
482
483     client.vw.provision "client_ntp",
484         type: "shell",
485         provision_order: true,
486         path: "${provision/client/ntp.sh}"
487
488     client.vw.provision "client_ntp",
489         type: "shell",
490         provision_order: true,
491         path: "${provision/client/ntp.sh}"
492
493     client.vw.provision "client_ntp",
494         type: "shell",
495         provision_order: true,
496         path: "${provision/client/ntp.sh}"
497
498     client.vw.provision "client_ntp",

```

Рис. 18: Настройка Vagrantfile

В результате выполнения лабораторной работы были получены навыки использования журналов системных событий